

B.E. Even Semester Midterm Examination, 2024-2025

Subject: Operating System

Paper Code: PCC-IT 403

Full Marks: 15 | **Time:** 1 Hr.

Q1. What is a system call? (2)

Answer:

A system call provides the programmatic interface between a user-space application and the services provided by the operating system's kernel. Since user applications are restricted from directly accessing hardware or critical system functions (for security and stability), they must execute a system call to request the OS to perform tasks like file I/O, process creation, or memory allocation on their behalf.

Q2. What does a short-term scheduler do? (2)

Answer:

The short-term scheduler (also known as the CPU scheduler) is responsible for selecting one of the processes from the **ready queue** (processes that are loaded in memory and ready to execute) and allocating the CPU to it. It executes very frequently (every few milliseconds) to ensure responsive multitasking and efficient CPU utilization.

Q3. Calculate the waiting time for process C when three processes- Process A, Process B and Process C are coming in sequence with CPU burst time 10, 20 and 30 seconds respectively, and FCFS CPU scheduling algorithm is followed. (2)

Answer:

First-Come-First-Serve (FCFS) schedules processes strictly in the order they arrive. Assuming they arrive in sequence at time 0:

- **Process A:** Arrives first, executes immediately. Waiting time = sec.
(Executes from to).
- **Process B:** Must wait for A to finish. Waiting time = sec.
(Executes from to).
- **Process C:** Must wait for both A and B to finish.
Waiting time = Burst time of A + Burst time of B = seconds.
Answer: The waiting time for Process C is **30 seconds**.

Q4. State the forms of memory wastage in fixed and dynamic partitioning schemes. (2)

Answer:

- **Fixed Partitioning:** Leads to **Internal Fragmentation**. This happens when a process is allocated a fixed-size partition that is larger than the process itself. The remaining unused space inside that partition is wasted.
- **Dynamic Partitioning:** Leads to **External Fragmentation**. As processes are loaded and removed, free memory is broken into small, non-contiguous blocks. Even if total free memory is enough for a new process, it cannot be allocated because the memory is not contiguous.

Q5. Calculate the effective memory access time for a paging system with 90% TLB hit where main memory access time is 20 nano second and TLB access time is 2 nano second. (2)

Answer:

Given:

- Hit Ratio () = 90% =
- Miss Ratio () = 10% =
- TLB Access Time () = ns
- Main Memory Access Time () = ns

Formula for Effective Memory Access Time (EMAT):

(Note: A miss requires 2 memory accesses: 1 for the page table, 1 for the actual data).

ns.

Answer: The effective memory access time is **24 nano seconds**.

Q6. What is demand segmentation? (2)

Answer:

Demand segmentation is a memory management strategy analogous to demand paging. Instead of loading an entire process into main memory at once, only the necessary segments (logical units of a program like main function, arrays, libraries) are loaded into memory when they are explicitly referenced or "demanded" during execution.

Q7. Explain, in brief, the role of Page Map Table Base Register (PMTBR) and Page Map Table Limit Register (PMTLR) in paging memory management scheme. (3)

Answer:

When an OS uses a paging scheme, page tables are stored in main memory. To access them efficiently:

- **PMTBR (Page Map Table Base Register):** This register points to the base (starting physical address) of the page table for the currently running process in the main memory. It allows the hardware MMU to quickly locate the page table to translate logical addresses to physical addresses.
- **PMTLR (Page Map Table Limit Register):** This register holds the total size (number of entries) of the process's page table. It is used to protect memory by checking if the requested logical page number falls within the valid range of the process. If a process tries to access a page number greater than or equal to the PMTLR, a trap (segmentation fault) is generated.